Messengers in High-Energy Astrophysics

(Lecture 2)

Satyendra Thoudam
Khalifa University

Outline of this lecture

- Introduction: Units and conventions.
- Galactic Coordinate system.
- Messengers in high-energy astrophysics.
- Interaction of high-energy cosmic messengers
- Potential sources of cosmic rays.
- Why multi-messenger?

Astronomical conventions and units in high-energy astrophysics

Distances

- Unit: parsec (pc).
- 1 pc = 3 x 10^{16} m = 3.26 light years.
- 1 kpc = 1000 pc.

Direction

- Unit: arc sec to degrees
- Will commonly use degrees for this course

Flux

- Unit: 1/(cm^2 s sr eV)
- 1/(cm^2 s sr GeV)
- 1/(m^2 s sr GeV)
- 1/(cm^2 s sr TeV)

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Energy

- Unit: electronvolts (eV).
- 1 KeV = 10^3 eV
- 1 MeV = 10^6 eV
- 1 GeV = 10^9 eV
- 1 TeV = 10^{12} eV
- 1 PeV = 10^{15} eV
- 1 EeV = 10^{18} eV

Mass of objects

- Unit: Solar mass ($M_\odot$)
- 1 $M_\odot$ ~ 2 x 10^{30} kg
- Mass of jupiter ~ 0.001 $M_\odot$
- Mass of our Galaxy ~ 10^{11} $M_\odot$

Electric charge

- electron: 1.6 x 10^{-19} C
- proton: 1.6 x 10^{-19} C
- nuclei: Z x charge of a proton

Mass of particles

- electron: 9.1 x 10^{-31} kg
- proton: 1.67 x 10^{-27} kg
- nuclei: Z x mass of a proton

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Other quantities

- Age of an object: 10^{xx} years
- Matter/energy density: #/cm^3
- Magnetic field: gauss (G), micro-gauss (μG), nano-gauss (nG)

Astronomical conventions and units in high-energy astrophysics

Commonly used conversion factors

(proton rest mass) $\times\, c^2 = m_{\rm p} c^2 = 938.3$ MeV $\approx 10^9$ eV $= 1$ GeV ,

(electron rest mass) $\times\, c^2 = m_{\rm e}\, c^2 = 0.511$ MeV $\approx 5 \times 10^5\, eV = 0.5$ MeV ,

1 eV $= 1.602 \times 10^{-19}\, J = 1.602 \times 10^{-12}$ erg .

1 kpc = 3 x 10^{21} cm

1 year = 365 x 24 x 60 x 60 seconds

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Some conventions used for particles

- Energy per nucleon: *E/n*
- Momentum per nucleon: *p/n*
- Distance traversed by a particle (path length): g/cm^2

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Some relativistic relations

$$\text{total energy} = (\text{rest energy} + \text{kinetic energy}) = \gamma m_0 c^2 ,$$

$$\text{kinetic energy} = \text{total energy} - \text{rest mass energy} = (\gamma - 1) m_0 c^2 ,$$

$$\text{Lorentz factor} \quad \gamma = (1 - V^2/c^2)^{-1/2} , \quad \beta = V/c$$

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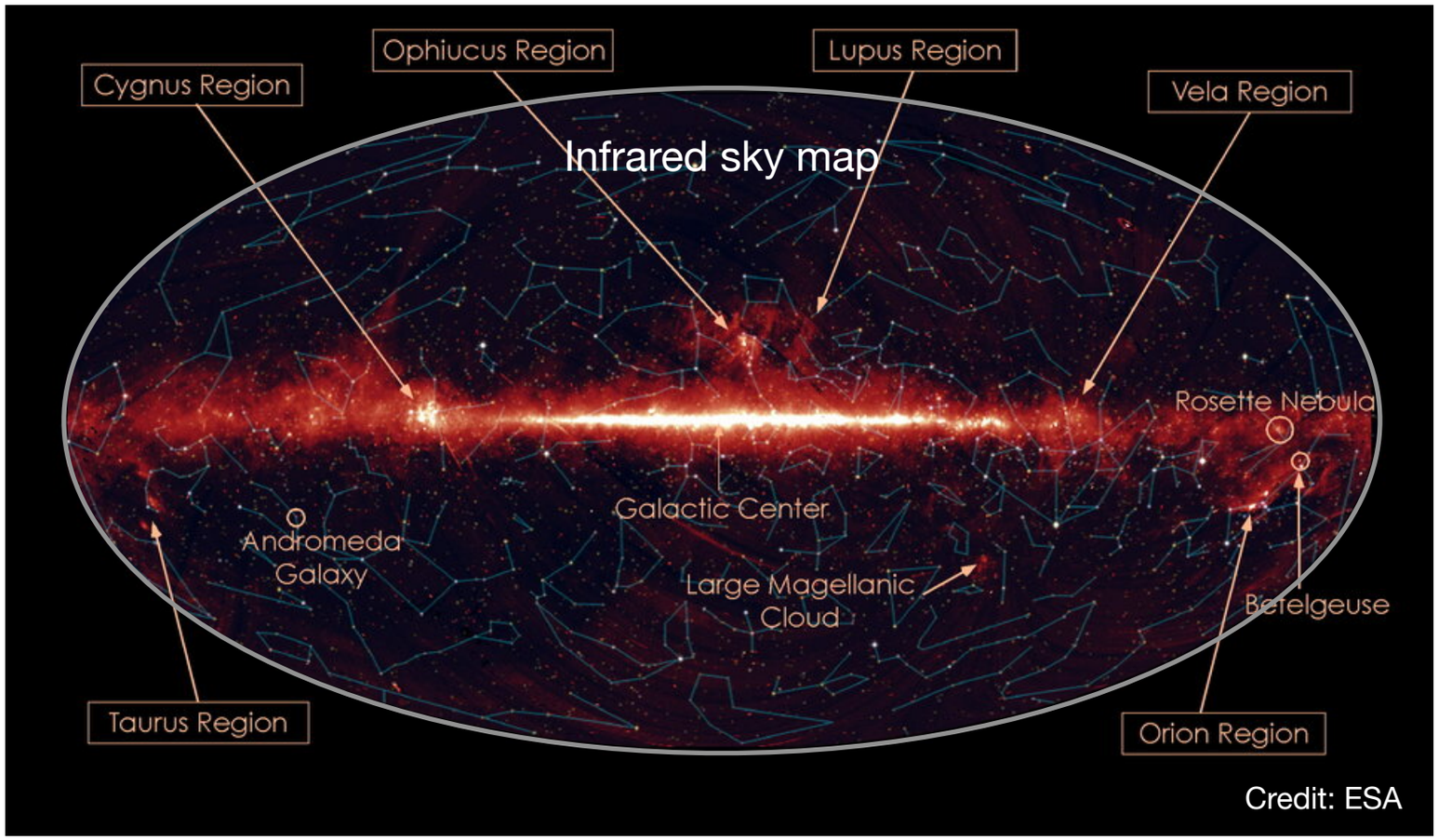
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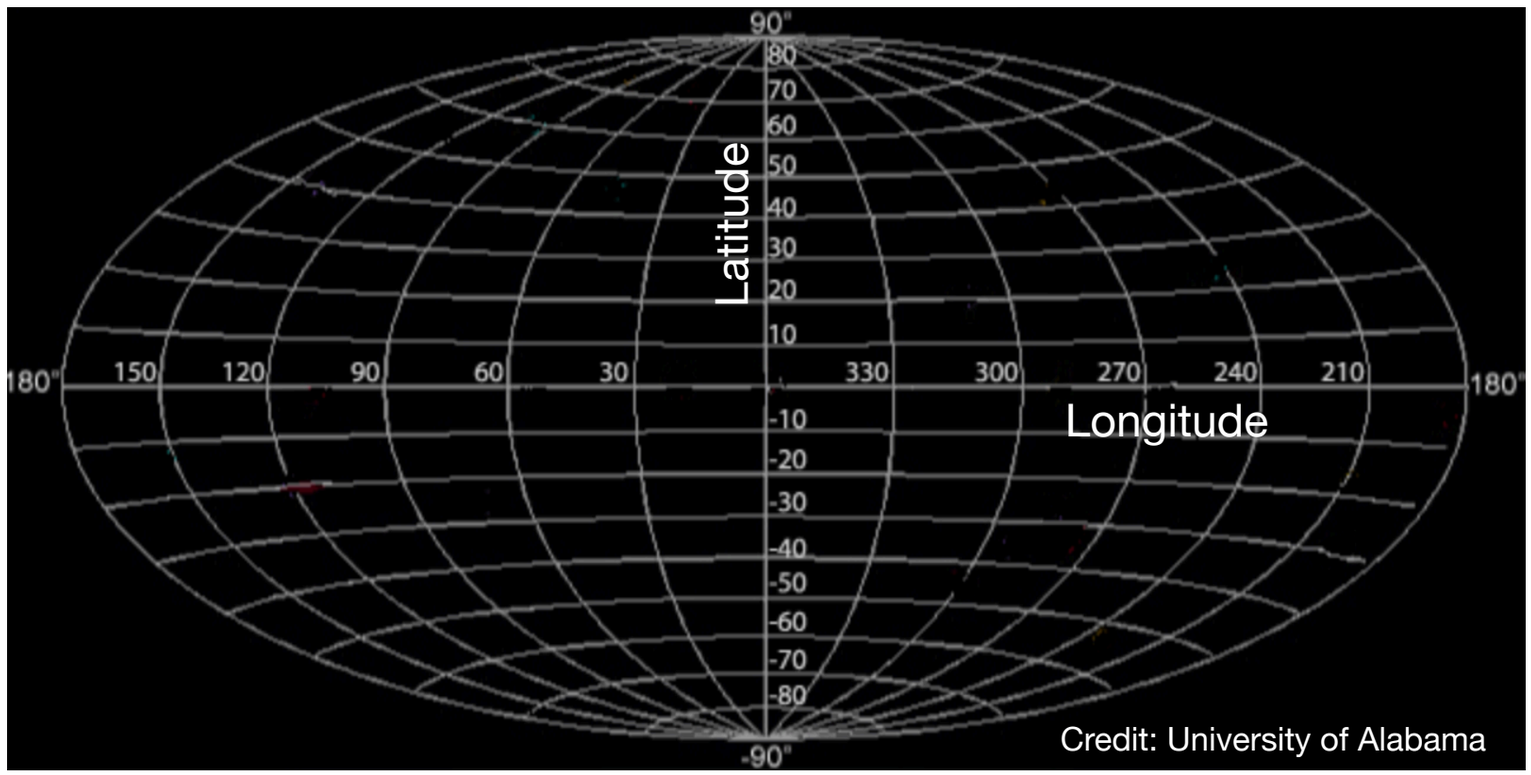
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*** Exercise: At what kinetic energy (GeV) does a proton has a velocity of 99.9999%c ?**

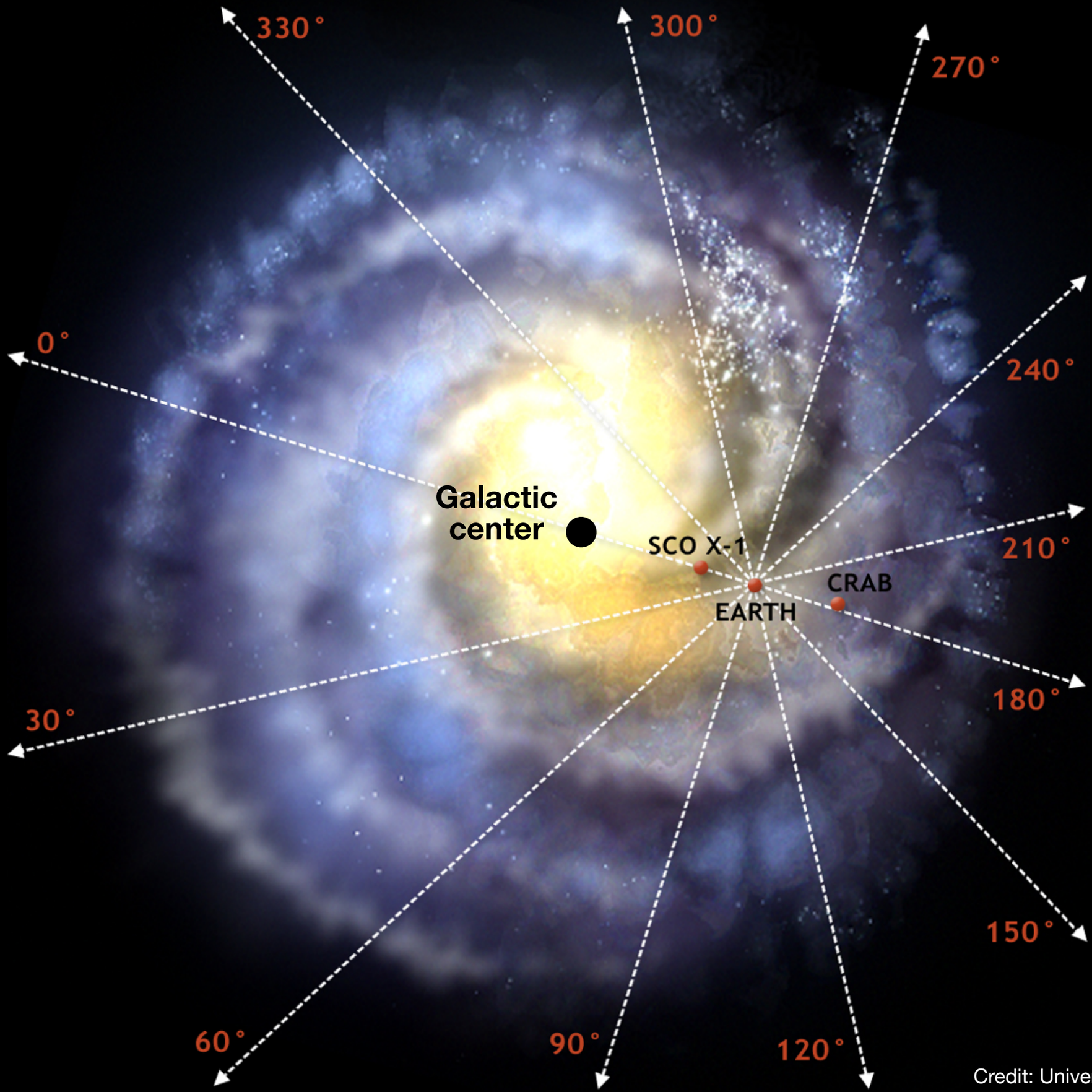
*** What is the corresponding energy for an electron?**

Galactic coordinate system



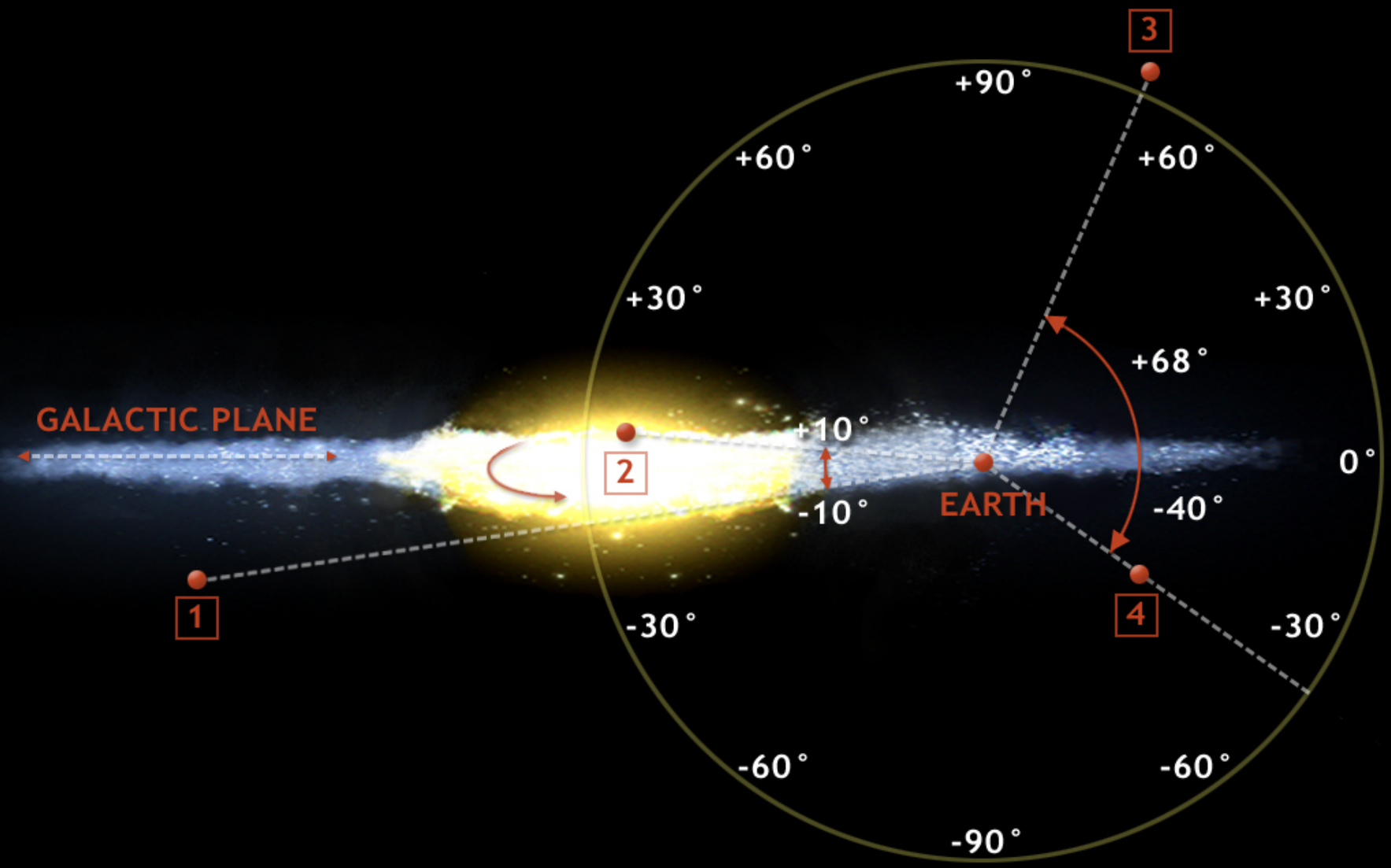


GALACTIC COORDINATE SYSTEM: DEGREES LONGITUDE



Credit: University of Alabama

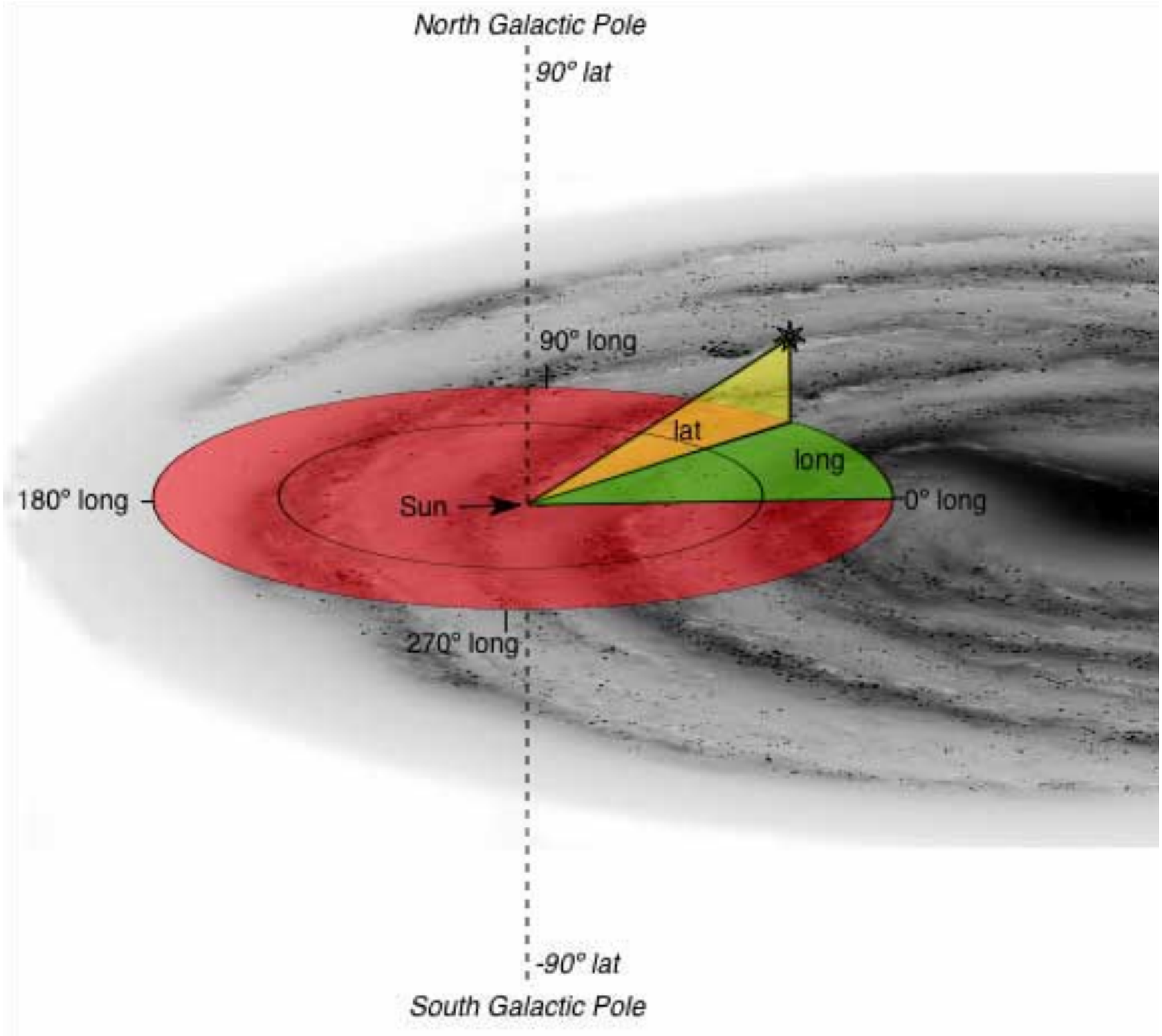
GALACTIC COORDINATE SYSTEM: DEGREES LATITUDE



CELESTIAL OBJECT DIRECTION OF GALACTIC ROTATION

Credit: University of Alabama

Galactic coordinate system



Credit: University of Alabama

High-Energy Astrophysics

•Astrophysics of high-energy processes and their application in astrophysical and cosmological contexts.

•High-energy processes:

- Formation of massive compact objects like black holes.
- Acceleration of charged particles to extremely high energies.
- The origins of enormous fluxes of high-energy particles and radiation from astrophysical objects.
- The physical processes in the interiors and environments of neutron stars.
- The nature of the dark matter.
- The origin and the expected fluxes of gravitational waves in extreme astrophysical environments.

•High-energy astrophysics makes feasible the study of the properties of matter and physical properties in extreme astrophysical environments.

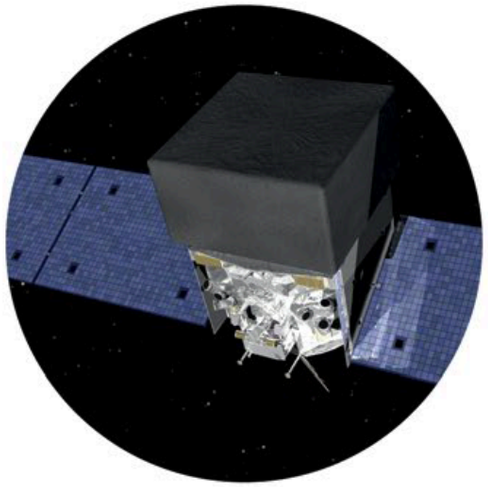
We want to study the COSMOS (our home)

Different types of cosmic messengers in high-energy astrophysics

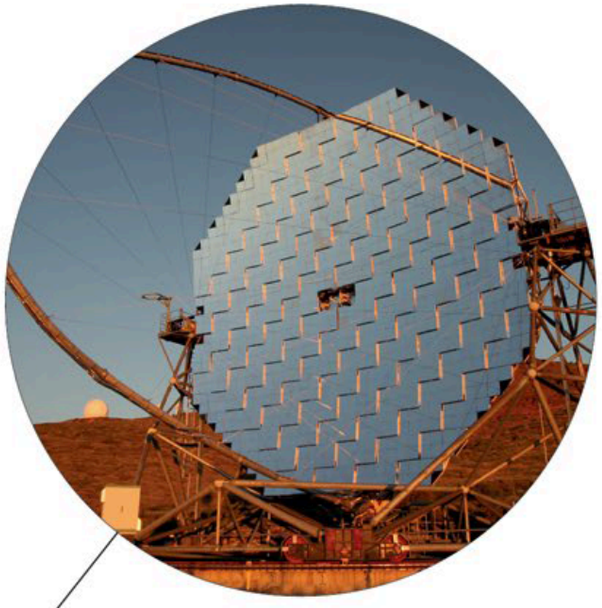
Gravitational waves

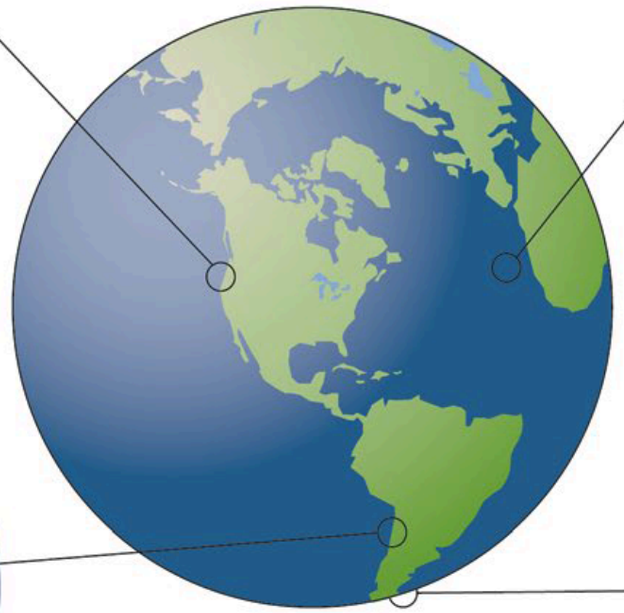


Gamma rays



VHE gamma rays







Cosmic rays



Neutrinos

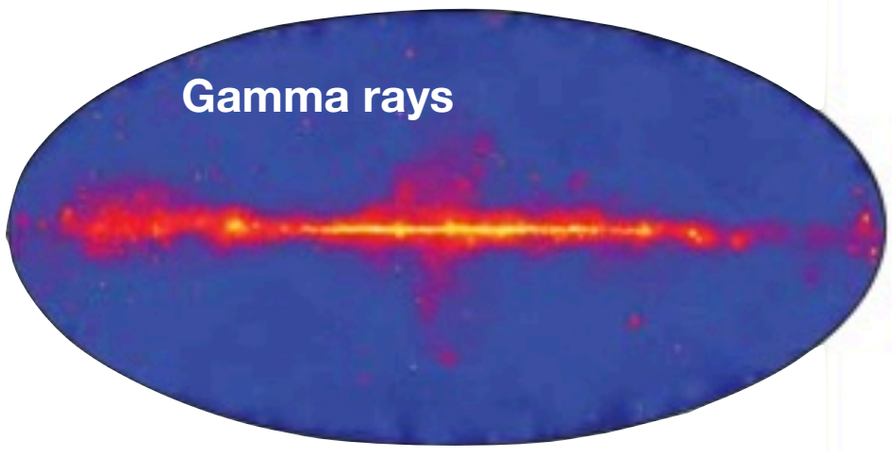
Meszaros+ 2019, Nature Reviews

Each messenger gives a unique view of the Universe

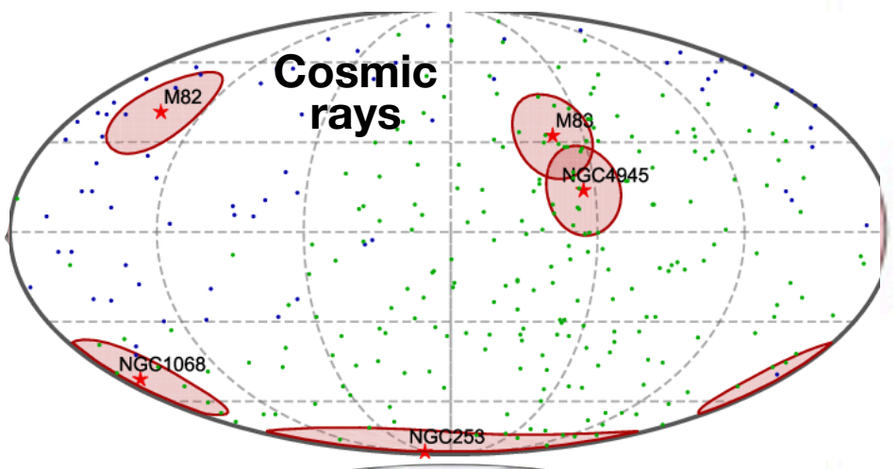
Visible: One of the most comprehensive maps of the visible sky ever made, this familiar view of the Milky Way Galaxy comes from ESA's Gaia spacecraft.



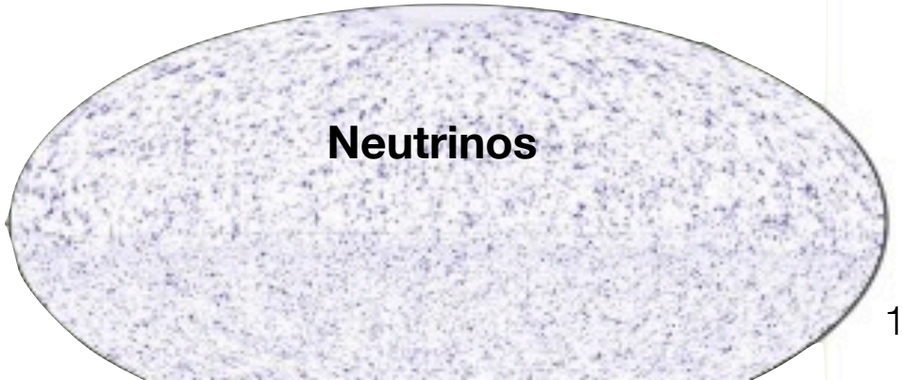
Gamma rays: This view from NASA's Fermi space telescope reveals gamma-ray sources ranging from pulsars in our own galaxy to distant blazars. The galactic disk appears flat due to the use of galactic coordinates.



Cosmic rays: This map of the cosmic ray sky, produced by IceCube and the High Altitude Water Cherenkov Observatory, is less detailed but shows that more ultrahigh-energy cosmic rays are coming from one side of the sky (shown in red) than the other (in blue). Astronomers aren't yet sure what causes this imbalance.



Neutrinos: IceCube's all-sky neutrino map is also in its early stages of being collected, and individual sources are only on the verge of detection.



Multi-messenger Astronomy/Astrophysics:

- Cosmic rays
- Photons
- Neutrinos
- Gravitational waves



Gravitational waves: The plotted areas are astronomers' best estimate of the origins of LIGO/Virgo's first 11 gravitational wave detections. When additional gravitational facilities come online, they should help pinpoint future sources with greater precision.

Credit: astronomy.com

Interaction processes of high-energy cosmic messengers

Cosmic rays
(p, He, …, Fe, e±)

Photons
(Gamma rays)

Neutrinos
(Astrophysical)

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Nuclei
(p, He, …, Fe)
+
matter/radiation/
magnetic field

Electrons
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Nuclei + Matter:

- Ionization/excitation, coulomb, Cherenkov
- Nuclear interactions:
  Pion production, disintegration (heavy nuclei)

Nuclei + Radiation:

- protons: Photopion, e± pair creation
- Heavy nuclei: Photo-disintegration

Nuclei + Magnetic field:

- Deflection/scattering

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Electrons
(e±)
+
Matter/radiation/
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Electron + Matter:
- Ionization/excitation, coulomb
- Cherenkov emission
- Bremsstrahlung
- annihilation (for e+)

Electron + Radiation:
- Inverse Compton

Electron + Magnetic field:
- Synchrotron
- Scattering

Interaction processes of high-energy cosmic messengers

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(p, He, …, Fe, e±)

Photons
(Gamma rays)

Neutrinos
(Astrophysical)

Low/High energy γ-rays

+

Interstellar matter/photon

Low energy γ-rays + Matter:

- Photoelectric effect

- Compton scattering

- e± pair creation

High energy γ-rays + Photons:

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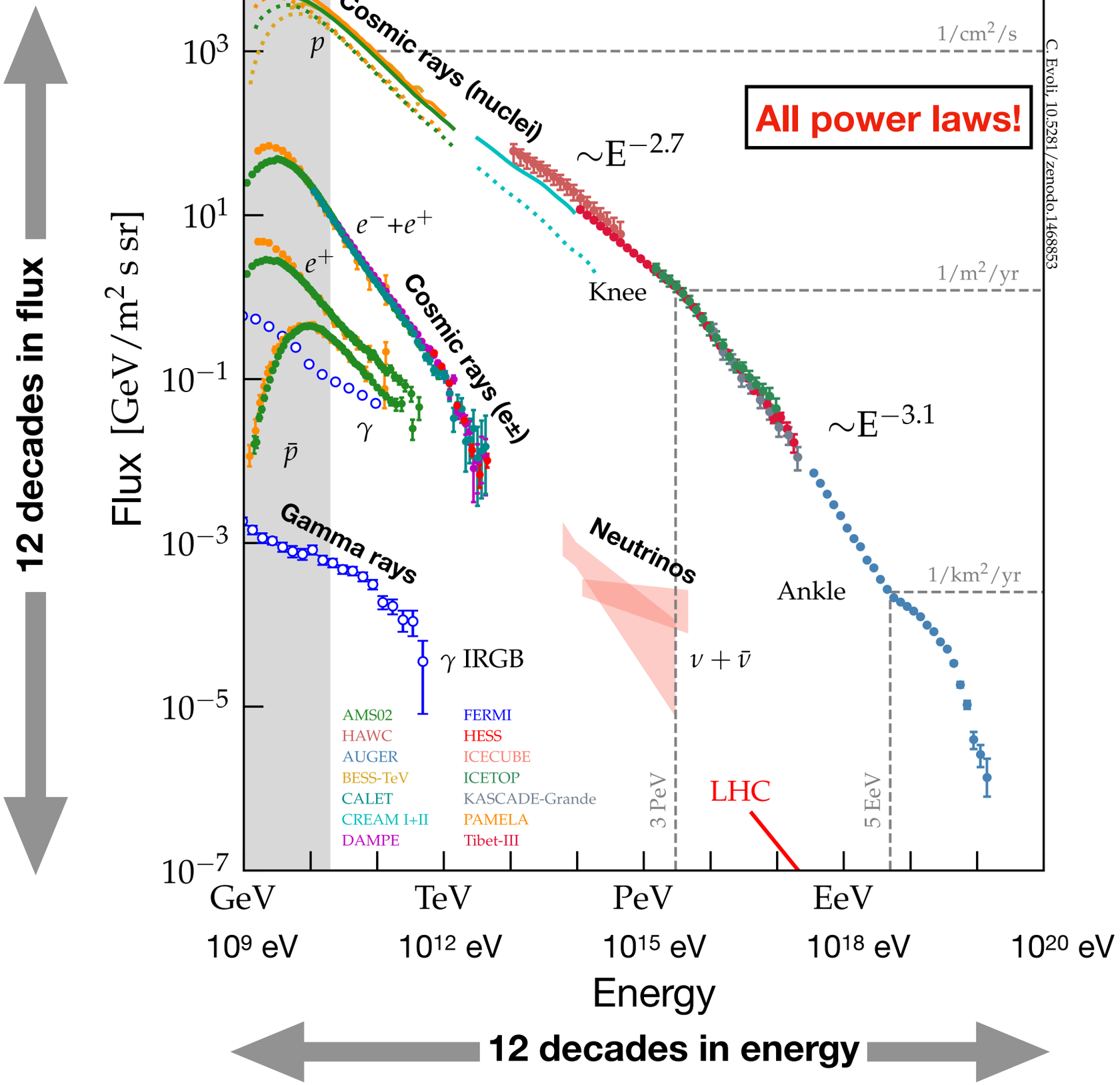
High energy γ-rays + Photons:
- e± pair creation

Neutrinos
(Astrophysical)

Neutrinos barely interact.

Neutrinos + Matter:
- produces electron (for ν_e)
- produces muon (for ν_μ)

Fluxes of different messengers observed at the Earth



The big question in Astroparticle physics

What/where are the sources of cosmic rays (nuclei)?

- Cosmic rays are scattered/deflected by magnetic fields. So, they cannot point back to their sources.

- Several properties of cosmic rays measured over the last 100 years. A number of theoretical models for their acceleration and propagation have been developed. Still their origin remains unclear.

- Gamma rays and neutrinos are neutral. So, they can pin-point their sources.
  The problem is: neutrinos are hard to detect, and gammas are produced both by cosmic-ray nuclei and electrons.

The big question in Astroparticle physics

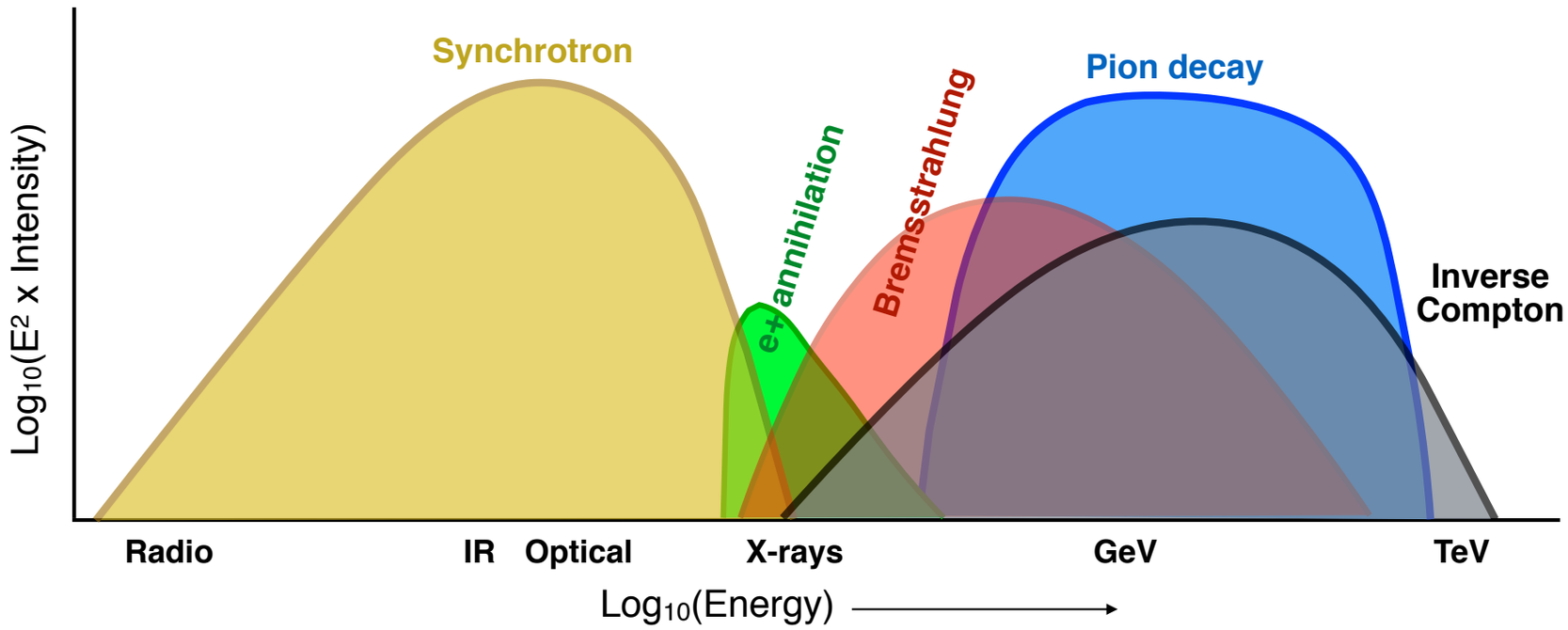
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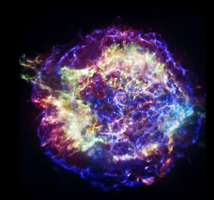
Non-thermal emission from an E^{-2} cosmic-ray (proton & electron) spectrum



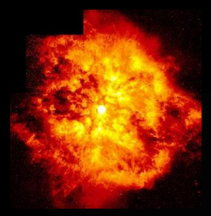
Potential sources of cosmic rays from γ-ray observations

Galactic sources (Mostly in Galactic plane)

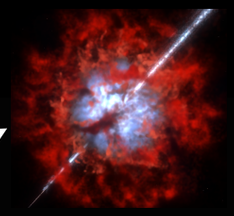
Extragalactic sources (Mostly at higher latitudes)



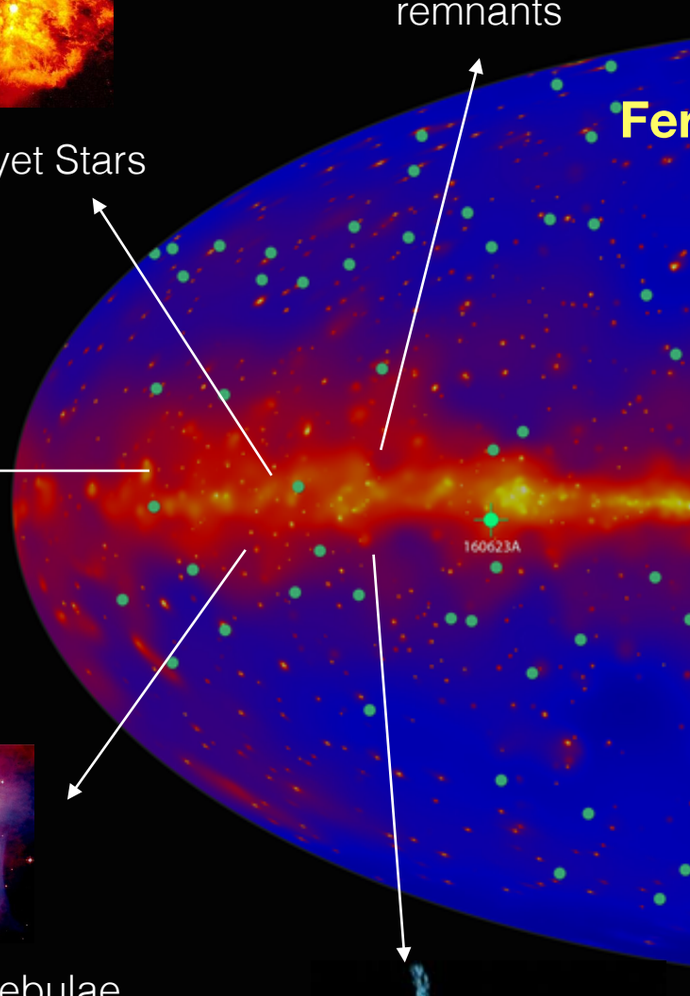
Supernova remnants



Wolf-Rayet Stars

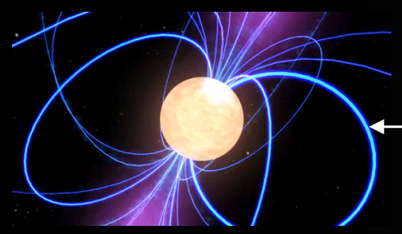


Gamma-ray bursts



Fermi γ-ray sky map (>10 GeV)

Diffuse γ-rays

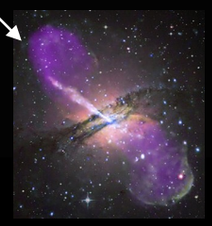


Pulsars

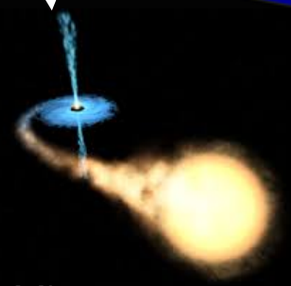


Pulsar Wind Nebulae

Fermi Bubbles



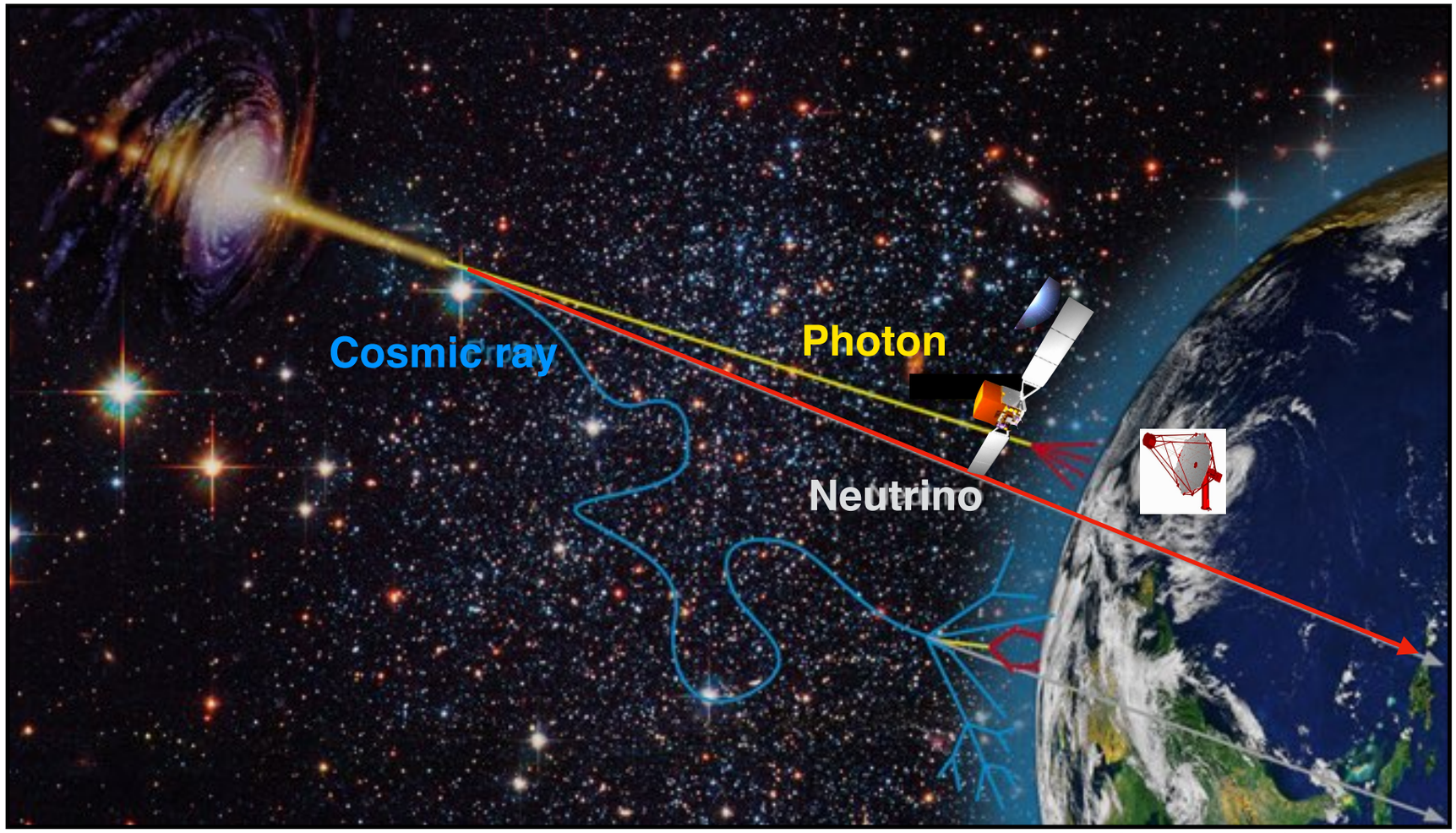
Active galactic nuclei



Microquasars

Credit: NASA/Fermi Collaboration/ESA

Multi-messenger Astrophysics: Cosmic rays, gamma rays & neutrinos



Credit: https://astro.desy.de

Cosmic-ray measurements

Properties of cosmic rays

- Spectrum (nuclei, e±)
- Composition
- Direction of sources (not really)

Gamma-ray measurements

Properties of the sources and the environment

- Spectrum
- Direction of sources

Neutrino measurements

Properties of the sources

- Spectrum
- Direction of sources

Next, we will learn how high-energy particles interact and produce radiation.